Structural Consistency for Confidence Estimation in LLM-Based Information Extraction: A Systematic Comparison and the Correlation-Selection Gap

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Abstract

Multi-sample generation with large language models (LLMs) is increasingly used for structured prediction tasks such as named entity recognition (NER) and relation extraction (RE), yet practitioners lack principled guidance on which confidence signals to trust—and for what purpose. We present the first systematic comparison of five unsupervised confidence signals across four families for LLM-based information extraction: model log-probability, surface-level agreement (Fleiss' Kappa, exact match), and two structural consistency measures (soft Jaccard similarity, entity-level majority voting). On two NER datasets (SciERC, CoNLL-2003) with two models (Qwen3-8B, LLaMA-3.1-8B), structural consistency signals provide the strongest instance-level quality estimation, with entity voting achieving AUROC 0.73 and ρ=0.50 on SciERC NER—far above log-probability (AUROC 0.56, ρ=0.14). However, we identify a *correlation-selection gap*: despite strong instance-level correlation, **no** unsupervised signal beats greedy decoding for NER best-of-N selection, while large oracle headroom (+16pp F1) exists. On RE, only log-probability enables marginal selection improvement (+0.7pp), while all signals degrade due to compound error propagation. We provide a practical decision framework: use structural consistency for selective prediction and abstention; use log-probability for best-of-N selection on tasks where greedy is suboptimal; and for closed-source APIs where log-probabilities are unavailable, structural consistency offers the only viable confidence signal.

1 Introduction

Test-time scaling (TTS)—generating multiple candidate outputs and selecting the best—has emerged as a powerful strategy for improving LLM reasoning Wang et al. (2023); Snell et al. (2024); Brown et al. (2024). For reasoning tasks with discrete answers, self-consistency via majority voting reliably improves accuracy. But for *structured prediction* tasks—named entity recognition (NER) and relation extraction (RE)—the outputs are *sets* of typed spans and relations, not single tokens. How should practitioners assess the quality of such structured outputs, and can multi-sample generation improve them?

Prior work on LLM confidence estimation has largely focused on free-form generation Kuhn et al. (2023); Manakul et al. (2023); Kadavath et al. (2022) or classification Guo et al. (2017); Desai and Durrett (2020). For structured IE, Nédellec and Xia (2026) recently studied consistency–accuracy correlations for entity and relation extraction, but used surface-level string matching and acknowledged this "overlooks semantically consistent outputs." Test-time scaling methods such as CISC Huang et al. (2025), Self-Certainty Best-of-N Zhang et al. (2026), and ReASC Li et al. (2026) target reasoning tasks with discrete answers, leaving structured prediction unexplored.

We address this gap with a systematic empirical comparison of five unsupervised confidence signals across four families for LLM-based IE (Figure 1). Our taxonomy spans: (1) *model confidence* (sequence log-probability), (2) *surface agreement* (Fleiss' Kappa, exact match), (3) *set-aware structural similarity* (soft Jaccard over entity sets), and (4) *entity-level majority voting confidence*. We evaluate each signal for two distinct purposes that prior work has conflated: *instance-level quality estimation* (predicting which inputs yield reliable outputs) and *per-sample selection* (choosing the best output among N candidates).

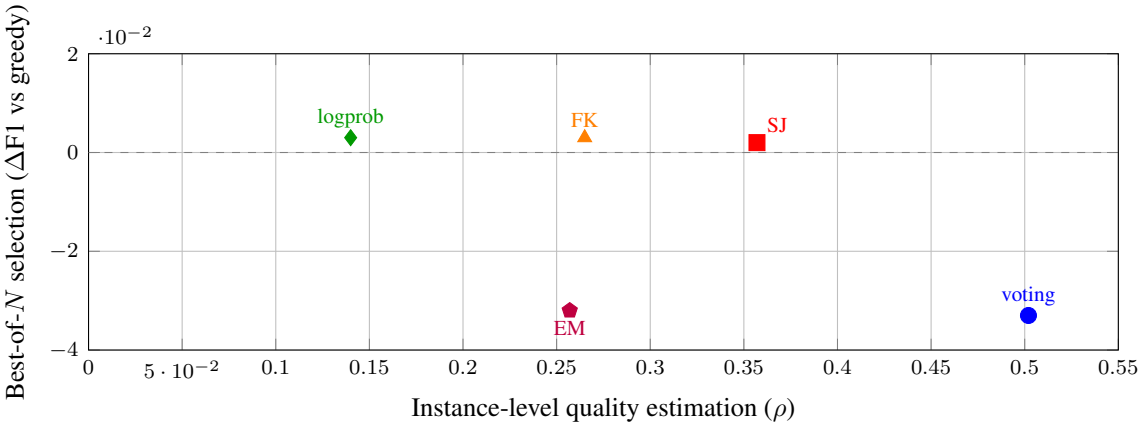


Figure 1: The correlation-selection gap on SciERC NER (N=16). x-axis: instance-level quality estimation (Spearman ρ with F1). y-axis: best-of-N selection gain over greedy. Signals with strong quality estimation (voting, SJ) fail at selection; the dashed line marks greedy performance. Oracle headroom is +16.9pp.

Our contributions are:

1. A systematic empirical comparison showing structural consistency signals (voting, soft Jaccard) are the strongest quality estimators for IE, significantly outperforming log-probability (ΔAUROC up to +0.17). The contribution lies in the taxonomy and comparative methodology, not in any individual signal.

2. The identification of a **correlation-selection gap**: strong instance-level quality estimation ($\rho \approx 0.50$) does not translate to effective best-of-N selection. We provide concrete practical guidance via a decision flowchart (Section 6).

3. An empirical analysis of task-complexity boundaries, showing compound error propagation in relational tasks degrades all confidence signals, with per-relation-type heterogeneity ranging from ρ=0.55 (HYPONYM-OF) to ρ=0.00 (FEATURE-OF).

2 Related Work

Uncertainty Quantification for LLMs. Uncertainty estimation in LLMs has been extensively surveyed Geng et al. (2025); Hou et al. (2025); Xiong et al. (2024); Yoshikawa et al. (2025). Approaches include token-level log-probabilities Kadavath et al. (2022), semantic entropy over meaning clusters Kuhn et al. (2023), semantic entropy probes for efficient estimation Kossen et al. (2024), and sampling-based methods such as SelfCheckGPT Manakul et al. (2023). For calibration, Guo et al. (2017) established that modern neural networks are poorly calibrated; Desai and Durrett (2020) and Kuleshov and Liang (2015) extended this to NLP. Our work differs by focusing on *structured set-valued* outputs where token-level confidence is insufficient.

Self-Consistency and Test-Time Scaling. Wang et al. (2023) showed majority voting over chain-of-thought samples improves reasoning accuracy. Recent extensions include CISC Huang et al. (2025) (confidence-weighted voting), Self-Certainty Best-of-N Zhang et al. (2026) (KL-divergence reward-free selection), ReASC Li et al. (2026) (reliability-aware adaptive consistency), Atomic Self-Consistency Chen et al. (2024) (sub-claim decomposition), Ranked Voting Liu et al. (2025), and Latent Self-Consistency Wu et al. (2025) (embedding-space agreement). Snell et al. (2024) and Brown et al. (2024) analyze TTS scaling laws. Verifier-based approaches Cobbe et al. (2021); Lightman et al. (2023) train reward models for

selection. All of these target reasoning/QA with discrete answers; we are the first to systematically compare confidence signals for *structured IE* and to identify the correlation-selection gap.

Confidence and Consistency for Structured IE. Nédellec and Xia (2026) study consistency–accuracy correlations for entity and relation extraction using surface-level string matching across four LLMs, acknowledging this approach "overlooks semantically consistent outputs." Our soft Jaccard metric addresses exactly this limitation. Culotta and McCallum (2004) estimated confidence for traditional CRF-based IE. Xiao and Wang (2019) and Shelmanov et al. (2021) study uncertainty in sequence labeling. Jagannatha and Yu (2022) address structured calibration. For ensemble/MC-dropout uncertainty, see Lakshminarayanan et al. (2017) and Gal and Ghahramani (2016). CoCoA Vashurin et al. (2025a) integrates confidence and consistency within an MBR framework for QA/summarization but does not address structured IE. Geifman and El-Yaniv (2017) formalize selective prediction, directly relevant to our quality estimation application.

3 Method

3.1 Problem Setup

Given an input x and a generative LLM, we sample N structured outputs $\{y_1, \ldots, y_N\}$ at temperature $T{>}0$. Each output y_i is a set of entities (for NER) or a set of (head, relation, tail) triples (for RE). We also obtain a greedy output y^* at $T{=}0$. The goal is twofold: (1) *quality estimation*: predict the quality (F1 score) of the greedy output y^* using information from the N stochastic samples; (2) *sample selection*: select the highest-quality sample from $\{y_1, \ldots, y_N\}$.

3.2 Confidence Signals

We evaluate five signals across four families:

(1) Log-probability (logprob). The normalized sequence log-probability of each sample: $s_{\text{lp}}(y_i) = \frac{1}{|y_i|}\sum_t \log p(y_{i,t} \mid y_{i,<t}, x)$. This is the only *per-sample* signal—it assigns different scores to different samples y_i, enabling direct best-of-N selection.

(2) Exact Match rate (EM). The fraction of sample pairs with identical string outputs: $s_{\text{EM}} = \binom{N}{2}^{-1}\sum_{i<j} \mathbb{1}[y_i = y_j]$. This is the direct analog of self-consistency voting Wang et al. (2023) applied to structured outputs.

(3) Fleiss' Kappa (FK). Inter-annotator agreement across N samples, treating each sample as an annotator. Computed at the entity level for NER and the triple level for RE.

(4) Soft Jaccard Similarity (SJ). A set-aware structural similarity that accounts for partial matches between entity sets. For two entity sets A and B, we compute an optimal matching via type-grouped Hungarian assignment, then:

$$\text{SJ}(A, B) = \frac{\sum_{(a,b)\in M} \text{overlap}(a, b)}{\max(|A|, |B|)} \tag{1}$$

where M is the matching and overlap measures span overlap within each entity type. The instance-level SJ score averages over all $\binom{N}{2}$ sample pairs. To avoid evaluation circularity, we use leave-one-out SJ: each sample's score is the mean SJ with the remaining $N{-}1$ samples (max $\Delta\rho = 0.0003$ vs full SJ).

(5) Entity Voting Confidence (voting_conf). For each entity mention extracted in any sample, we compute the fraction of N samples containing that entity (with fuzzy span matching). The instance-level score averages entity-level voting fractions weighted by entity frequency. Like SJ, this is an *instance-level aggregate*—it produces one score per input instance, not per sample.

Instance-level vs. Per-sample Signals. A critical distinction: signals (2)–(5) are *instance-level aggregates* that produce a single confidence score per input instance x. They predict *which inputs are easy/hard* but cannot rank individual samples y_i against each other. Only log-probability (1) is a *per-sample signal* that can rank candidates for selection. This distinction explains the correlation-selection gap we identify.

3.3 Evaluation Protocol

Quality Estimation. We measure rank correlation (Spearman ρ, Kendall τ) and discriminative ability (AUROC, with median F1 as threshold) between each signal and the greedy output's entity-level F1 score.

Sample Selection. We evaluate best-of-N selection: for each instance, select the sample with the highest signal score and report its F1. For instance-level signals, all samples from the same instance share the same score, so selection is equivalent to random choice—these signals *cannot* perform sample selection by design. We compare against greedy decoding (T=0), random selection, and oracle (best possible sample).

Conditional Evaluation. To separate model capability gaps from genuine uncertainty, we report results on the *conditional set*: instances where at least one sample achieves F1 > 0, excluding cases where the model systematically fails.

4 Experimental Setup

Datasets. We use SciERC Luan et al. (2018) (scientific text, 6 entity types, 7 relation types; n_{NER}=529, n_{RE}=397) and CoNLL-2003 Tjong Kim Sang and De Meulder (2003) (news text, 4 entity types; n=2,756). We also attempted WNUT-17 Derczynski et al. (2017) (social media NER) but abandoned it: 54% of samples had F1=0, indicating model capability failure rather than uncertainty.

Models. Qwen3-8B and LLaMA-3.1-8B, both instruction-tuned, used in a zero-shot structured extraction prompt. Sampling at T=1.0 with N=8 (primary) and N=16 (scaling analysis), three random seeds for robustness.

Evaluation. Entity-level micro-F1 as the ground-truth quality metric. We report Spearman ρ, Kendall τ, and AUROC for quality estimation; F1 for best-of-N selection.

5 Results

5.1 Quality Estimation: Structural Signals Dominate

Table 1 presents unified quality estimation results on SciERC. On NER, entity voting confidence achieves the highest correlation (ρ=0.502, AUROC=0.728), followed by soft Jaccard (ρ=0.357, AUROC=0.666). Both substantially outperform log-probability (ρ=0.140, AUROC=0.559), with the structural-vs-surface AUROC gap reaching 0.17 (voting vs logprob).

On RE, the pattern partially holds: voting confidence leads (ρ=0.422, AUROC=0.716), but log-probability (ρ=0.247, AUROC=0.635) outperforms SJ (ρ=0.211, AUROC=0.582) on the full set. This

Table 1: Unified quality estimation on SciERC (N=8, T=1.0). Spearman ρ, Kendall τ, and AUROC. Best in **bold**, second underlined.

	NER (n=529)			**RE** (n=397)		
Signal	ρ	τ	AUROC	ρ	τ	AUROC
voting_conf	**0.502**	**0.377**	**0.728**	**0.422**	**0.327**	**0.716**
SJ	0.357	0.266	0.666	0.211	0.162	0.582
FK	0.265	0.203	0.624	0.253	0.197	0.631
EM	0.257	0.192	0.623	0.100	0.084	0.504
logprob	0.140	0.101	0.559	0.247	0.181	0.635

Table 2: Quality estimation on CoNLL-2003 NER (Qwen3-8B, N=8, n=2,756).

Signal	ρ	τ	AUROC
EM	**0.451**	**0.400**	**0.816**
SJ	0.436	0.385	0.801
voting_conf	0.424	0.378	0.789
FK	0.400	0.357	0.774
logprob	0.316	0.247	0.753

reversal on RE is explained by compound error propagation (Section 5.7). On the conditional set (instances where at least one sample has F1>0), SJ recovers to ρ=0.368, substantially above logprob (ρ=0.146).

5.2 Cross-Dataset Generalization

Table 2 shows results on CoNLL-2003 NER. All structural signals outperform log-probability, with SJ achieving ρ=0.436 and AUROC=0.801 (vs logprob ρ=0.316, AUROC=0.753). EM performs strongest on CoNLL (ρ=0.451) due to the high baseline accuracy (greedy F1=0.912): at this performance level, exact string match has sufficient granularity.

5.3 Cross-Model Validation

Table 3 shows results with LLaMA-3.1-8B on SciERC, and Table 4 presents the full 2×2 cross-validation matrix. On NER, SJ consistently outperforms logprob across all four model×dataset configurations (4/4), with $\Delta\rho$ ranging from +0.10 to +0.22. On RE conditional set, the pattern holds even more strongly (LLaMA SJ ρ=0.405 vs logprob ρ=0.092).

5.4 The Correlation-Selection Gap

Table 5 and Figure 2 present our central negative finding. Despite strong quality estimation correlations, **no signal beats greedy decoding** for NER best-of-N selection. SJ selects outputs at F1=0.620 ($\Delta = +0.002$ vs greedy 0.618, within noise), while oracle performance is 0.787 (+16.9pp headroom). On RE, only log-probability achieves a marginal gain (0.410 vs greedy 0.403, +0.7pp), with oracle headroom of +22.8pp.

This gap arises because consistency signals are *instance-level aggregates*: they predict which *inputs* are reliable, not which *output* among N candidates is best. Log-probability, as the only per-sample signal, is the sole signal capable of ranking individual candidates—and it succeeds only on RE, where greedy performance is sufficiently suboptimal.

Table 3: Cross-model validation (LLaMA-3.1-8B, SciERC).

Subtask	Signal	ρ (full)	ρ (cond)
NER	SJ	**0.363**	0.311
	logprob	0.183	0.109
RE	SJ	**0.299**	**0.405**
	logprob	0.288	0.092

Table 4: 2×2 cross-validation matrix: NER SJ ρ (full set). SJ consistently outperforms logprob (lp) in all configurations.

	SciERC	CoNLL-2003
Qwen3-8B	0.357 / 0.140^{lp}	0.436 / 0.316^{lp}
LLaMA-3.1-8B	0.363 / 0.183^{lp}	0.431 / 0.307^{lp}

5.5 N-Scaling and Seed Robustness

Table 6 shows that increasing from N=8 to N=16 provides marginal improvement for NER SJ (ρ: $0.383 \rightarrow 0.398$) and is stable across three random seeds (mean ρ=0.412 ± 0.025). For RE, N=16 improves SJ from ρ=0.211 to 0.277, but still falls short of the 0.40 threshold achieved on NER. The multi-seed stability (SD=0.025) confirms our results are not artifacts of a particular random seed.

5.6 Ensemble Analysis

Table 7 examines whether combining signals improves performance. A logistic regression ensemble achieves NER AUROC of 0.735 (+0.7pp over best single signal voting_conf), but fails to improve RE AUROC (0.719 vs voting alone 0.716). The best linear combination is SJ+voting (α=0.3) for NER (ρ=0.489) and voting+logprob (α=0.6) for RE (ρ=0.429). The SJ-logprob Pearson correlation is 0.57 (NER) and 0.42 (RE), indicating moderate complementarity. Critically, even the ensemble *cannot improve NER best-of-N selection* (F1=0.604 vs greedy 0.609), reinforcing that the correlation-selection gap is fundamental, not a matter of signal strength.

5.7 Diagnostic Analysis: Why RE Is Harder

Table 8 reveals dramatic heterogeneity across relation types. SJ achieves ρ=0.545 on HYPONYM-OF but ρ=0.004 on FEATURE-OF. Fisher's Z-test confirms NER SJ ρ is significantly higher than RE SJ ρ (p=0.021), though bootstrap 95% CIs overlap (NER: [0.295, 0.459]; RE: [0.142, 0.339]).

Entity span errors account for 55.5% of RE false negatives, confirming compound error propagation: RE quality depends on correct entity extraction, and entity errors cascade into relation failures. This explains why all confidence signals degrade on the full RE set but recover on the conditional set (where baseline entity extraction succeeds).

5.8 Temperature Sensitivity

SJ is highly robust to temperature: $\Delta\rho = \pm 0.01$ between T=0.7 and T=1.0, while FK shifts by $\pm$0.06. T=1.0 preserves more oracle headroom (RE: 0.167 vs 0.131 at T=0.7) and diversity (1−SJ: 0.389 vs 0.313), though no current signal can exploit this headroom. Conditional filtering provides far larger improvements ($\Delta\rho = +0.07$ to $+0.14$) than temperature tuning—practitioners should prioritize filtering over temperature optimization.

Table 5: Best-of-N selection F1 (SciERC, N=16). Δ: difference vs greedy.

	NER		RE	
Signal	F1	Δ	F1	Δ
greedy	0.618	—	0.403	—
logprob	0.621	+0.003	**0.410**	**+0.007**
SJ	0.620	+0.002	0.397	−0.006
FK	0.621	+0.003	0.396	−0.007
voting_conf	0.585	−0.033	0.408	+0.005
random	0.586	−0.032	0.364	−0.039
oracle	0.787	+0.169	0.631	+0.228

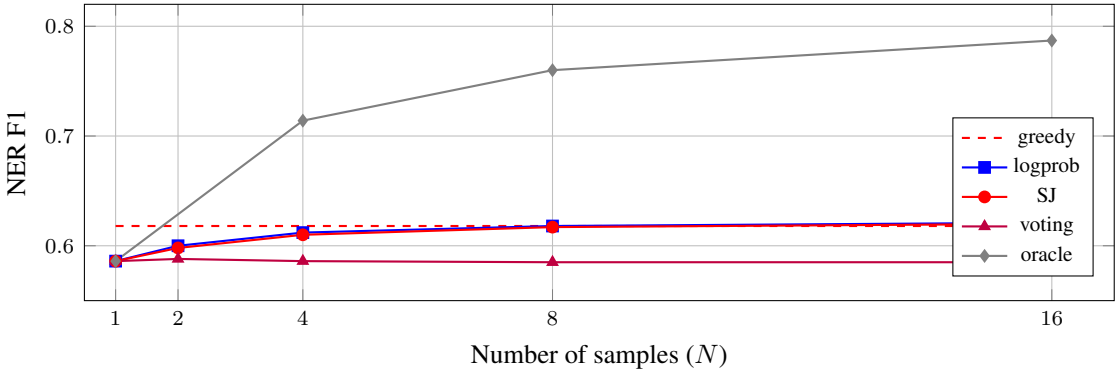


Figure 2: Best-of-N selection curves for SciERC NER. All unsupervised signals plateau near greedy performance (dashed red), while oracle grows monotonically, revealing a large unexploited headroom. Voting-based selection *hurts* performance due to its instance-level nature.

5.9 Capability Stratification

Stratifying instances by difficulty reveals that SJ is strongest on simple instances (entity count=0: ρ=0.689) and degrades on complex ones (entity count 3–5: ρ=0.353). On hard RE instances (greedy F1<0.25), SJ correlation drops to ρ=−0.043—effectively random. This confirms that consistency signals operate within the model's capability range and cannot rescue systematic failures. An independent proxy analysis (r=−0.021 between sentence length and F1) verifies that our difficulty strata are not circular with the evaluation metric.

5.10 OOD Robustness

In out-of-domain evaluation (Appendix Table 9), all signals degrade substantially (SJ: ρ drops from 0.357 to 0.15), but structural signals maintain positive correlation while log-probability inverts (ρ=−0.20). This has practical implications for closed-source API scenarios where log-probabilities are unavailable: structural consistency, despite weaker absolute correlation, provides directionally correct confidence estimates and does not actively mislead.

6 Practical Guidance

Based on our findings, we propose the following decision framework for practitioners:

When to use structural consistency (voting/SJ). For *selective prediction* and *abstention*: if the goal is to identify which instances are reliable (e.g., to route uncertain cases to human review), structural consistency provides the strongest signal (AUROC up to 0.73). Also preferred when: (a) log-probabilities

Table 6: N-scaling and seed robustness (SciERC, Qwen3-8B).

Config	NER SJ ρ	RE SJ ρ	NER AUROC	RE AUROC
N=8, s42	0.383	0.244	0.689	0.592
N=16, s42	0.398	0.277	0.697	—
N=16, s123	0.423	0.267	—	—
N=16, s456	0.414	0.268	—	—
N=16 mean±SD	0.412±0.025	0.271±0.006	—	—

Table 7: Ensemble analysis (SciERC, N=8).

	NER ρ	NER AUROC	RE ρ	RE AUROC
Best single (voting)	0.502	0.728	0.422	0.716
Best linear combo	0.489	—	0.429	—
Logistic ensemble	—	0.735	—	0.719

are unavailable (closed-source APIs), (b) domain shift is expected (structural signals degrade gracefully while logprob can invert), or (c) temperature robustness is needed.

When to use log-probability. For *best-of-N selection* on tasks where greedy decoding is suboptimal (e.g., RE). Log-probability is the only per-sample signal and the only one enabling effective sample selection in our experiments.

When neither helps. For NER best-of-N selection with current models, greedy decoding is near-optimal—multi-sample generation provides no selection benefit despite large oracle headroom. For tasks where the model systematically fails (e.g., social media NER with 54% zero-F1 rate), no confidence signal is informative.

Voting vs. SJ. Entity voting is simpler, faster, and empirically stronger on full sets. SJ generalizes to arbitrary structured outputs (relations, events) and is more temperature-robust. For standard NER, voting is the practical default; for novel structured tasks or when temperature stability matters, prefer SJ.

7 Discussion

The Correlation-Selection Gap as a General Phenomenon. Our finding that strong quality estimation does not imply effective selection may extend beyond IE. Instance-level consistency signals inherently measure *input difficulty*, not *output quality variation* within a fixed input. For tasks where within-input quality variance is low relative to between-input variance, the gap will be large. Developing per-sample structural signals—not just instance-level aggregates—is a key direction for closing this gap.

Applicability to Closed-Source APIs. Many production IE systems use closed-source APIs (GPT-4, Claude) that do not expose log-probabilities. In this setting, structural consistency signals offer unique value: they require only the generated text, not model internals. Our OOD results (Section 5.10) further suggest that log-probabilities can become *anti-correlated* with quality under domain shift, making structural signals the safer default.

Limitations. (1) We evaluate only 8B-parameter models; larger models may exhibit different patterns. (2) Both models are instruction-tuned; base models or fine-tuned models may differ. (3) We test on well-structured text domains (scientific, news); informal text (social media) remains unexplored—WNUT-17

Table 8: Per-relation-type quality estimation (SciERC RE, Qwen3-8B). Large heterogeneity: ρ_{SJ} ranges from 0.545 to 0.004.

Relation	n	ρ_{SJ}	ρ_{FK}	ρ_{VC}	Mean F1
HYPONYM-OF	36	**0.545**	0.646	0.628	0.513
CONJUNCTION	86	0.383	0.350	0.484	0.480
USED-FOR	299	0.236	0.255	0.286	0.422
PART-OF	44	0.239	0.348	0.283	0.297
EVALUATE-FOR	62	0.172	0.255	0.226	0.365
COMPARE	35	0.168	0.184	0.180	0.453
FEATURE-OF	50	0.004	0.052	0.014	0.254

was abandoned due to model capability failure. (4) RE evaluation is limited to SciERC; additional relational datasets would strengthen generalization claims. (5) We do not compare against supervised verifiers or reward models, which represent an upper bound on selection quality.

8 Conclusion

We present the first systematic comparison of unsupervised confidence signals for LLM-based information extraction, evaluating five signals across two tasks, two datasets, and two models. Structural consistency signals (entity voting, soft Jaccard) provide the strongest instance-level quality estimation, substantially outperforming log-probability. However, we identify a fundamental correlation-selection gap: no unsupervised signal beats greedy decoding for NER best-of-N selection despite +16pp oracle headroom. On relational tasks, compound error propagation degrades all signals, with dramatic per-relation heterogeneity. We provide practical guidance distinguishing when to use structural consistency (selective prediction, closed-source APIs) vs. log-probability (sample selection on hard tasks), and characterize the task-complexity boundaries within which confidence estimation is reliable.

A OOD Robustness Details

Table 9: Out-of-domain robustness. Structural signals degrade but remain positive; log-probability inverts.

Signal	In-Domain ρ	OOD ρ	Δ
SJ	0.357	0.15	-0.207
FK	0.265	0.10	-0.165
logprob	0.140	-0.20	-0.340

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